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ABSTRACT

Fluid viscosity plays an important role in multiphase flows, and the many-body dissipative particle dynamics (MDPD) method is an efficient means of simulating such flows at the mesoscopic scale. As the viscosity of the standard MDPD (S-MDPD) fluid cannot be efficiently adjusted, a transverse MDPD (T-MDPD) scheme is newly proposed to tune the viscosity of an MDPD fluid over a large range. With a lateral friction coefficient added to the S-MDPD form, the viscosity of the T-MDPD fluid is higher than that of an S-MDPD fluid, and the viscosity is about five times larger than that of an S-MDPD fluid with a friction coefficient of 40.5. In a T-MDPD fluid, the viscosity is much more sensitive to the new transverse friction coefficient, as it increases about five times higher when this coefficient increases from 4.5 to 40.5, while the viscosity only increases two times higher with the same variation of the original coefficient. By increasing the repulsive coefficient, the liquid particle number density, or the cutoff radius, the viscosity of the T-MDPD fluid is enhanced as well. Based on this extension of the MDPD scheme, a quantitative expression for the variation of the viscosity in the current T-MDPD fluid is derived. In future simulations of multiphase flows using an MDPD scheme, the transverse effect can be extended to effectively tune the viscosity, and this empirical expression will be useful to predict the viscosity of the T-MDPD fluid.

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I. INTRODUCTION

In microfluid dynamics, molecular dynamics (MD) simulations are widely applied to capture the underlying physical phenomena and investigate the physical mechanisms in detail.^{1–5} As MD simulations are based on all atoms in a system, the computational cost is considerable, and so such simulations are effectively restricted to nanodroplets or nanobubbles. The dissipative particle dynamics (DPD) method, in which a single coarse-grained particle in a DPD system represents an entire cluster of molecules, has been developed to overcome the restrictions of MD simulations, as it is computationally faster than the corresponding atomistic-based approach.^{6–9}

The classical DPD model, which only considers the momentum equation governing the evolution of the flow field, is incapable of modeling certain problems, such as liquid–vapor coexistence in one-component multiphase fluid systems, heat flow in non-isothermal systems, and the diffusion-reaction process in biological systems. To this end, several extensions of the DPD method, such as the many-body DPD model,^{10–12} energy-conserving DPD model,^{13–15} and transport

DPD model,^{16,17} have been developed in recent years to tackle the challenges faced by the classical DPD form.

Among the extensions of DPD methods, the many-body DPD (MDPD) model has been widely used in simulations of multiphase flows, such as for determining contact angles in capillary flow, examining liquid droplets on solid surfaces, and studying bubble formation processes.^{18–23} In multiphase flows, the viscosity plays a significant role in determining the fluid properties.^{24–29} For example, Kim and coworkers found that, as the viscosity of an infused silicone oil layer decreased, the droplet impact dynamics (i.e., spreading rate and maximum spreading diameter) on lubricant-infused surfaces would change.³⁰ Moreover, by studying the turbulent flow over a lubricated surface, Chang *et al.* found that the drag reduction is highly dependent on the lubricant viscosity, and a decrease in the viscosity improves the drag reduction effect.²⁶ Additionally, the viscosity is essential for mapping the DPD units, such as length, mass, and time, to a physical system.^{19,31}

In recent decades, many simulations have been conducted to find the viscosity in an MDPD model.^{32–35} Recently, Xiao *et al.* employed

both Poiseuille and Couette flow methods to measure the viscosity in an MDPD formulation. They found that the results obtained by these two nonequilibrium methods are consistent with each other and identified a mapping for the viscosity of MDPD models with different numerical parameters based on Poiseuille flows.³³ By modifying the equation of state for the MDPD system, Zhao *et al.* also extended the MDPD model and investigated the dynamics of non-Newtonian droplets, allowing them to explore the influence of the repulsive coefficient, density, and cutoff radius on the viscosity of the MDPD fluid.³⁴

However, as the pairwise forces in a DPD/MDPD equation only act on the relative velocity along the interparticle axis, the DPD/MDPD thermostat cannot effectively control liquid properties such as the viscosity and diffusion constant.³⁶ For DPD simulations, there have been several attempts to extend the equations and tune these quantities by changing the parameters of the DPD thermostat. For instance, Christoph *et al.* exploited a damping of the perpendicular components of the relative particle velocities in a DPD system and found that the viscosity increases sharply with increasing perpendicular friction coefficient in this transverse DPD scheme, while the viscosity only changes slightly as the parallel friction coefficient varies in a standard DPD fluid.³⁷ Recently, by including the lateral dissipative force in addition to the central ones in the standard DPD form, Takayuki *et al.* showed that this transverse DPD scheme can effectively tune the dynamic properties of nanometer-thick liquid films and found that the viscosity is much more sensitive than in a standard DPD fluid.³⁸

Although previous research has shown that the viscosity of the DPD modeled fluid is sensitive to the transverse friction coefficient, there has been little investigation into the transverse effect on the fluid dynamic properties in an MDPD model. In the current work, by adding a lateral coefficient that runs perpendicular to the parallel direction in standard MDPD (S-MDPD) simulations, a transverse MDPD (T-MDPD) scheme is used to explore the viscosity of an MDPD fluid. The results show that the viscosity of the T-MDPD fluid is much more sensitive to the friction coefficient and has a much greater magnitude than the viscosity of an S-MDPD fluid. As well as the friction coefficient, the other conservative force parameters are considered in analyzing the viscosity of the T-MDPD fluid, and an expression for predicting the viscosity is derived. This T-MDPD scheme can be used to effectively tune the fluid viscosity, allowing the droplet impact behaviors on solid or liquid surfaces to be adjusted, the flow slip characteristics over liquid–gas or liquid–liquid interfaces to be modified, and other applications in multiphase flows to be examined in greater detail.

The remainder of this paper is organized as follows. Section II describes the simulation method, which consists of the MDPD scheme, simulation setup, and parametrization. The variation of the viscosity is then discussed for various coefficients and an empirical expression is derived for a T-MDPD fluid in Sec. III. Finally, a summary of this study is presented in Sec. IV.

II. SIMULATION METHODS

A. MDPD scheme

In MDPD simulations, the particle interacts with three pairwise forces, and the governing equation of the i th particle is given by

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = \sum_{j \neq i} \mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R, \quad (1)$$

where $\mathbf{r}_i$ is the position of particle i , m_i is the mass of particle i , $\mathbf{F}_{ij}^C$ denotes the conservative force on the i th particle, $\mathbf{F}_{ij}^D$ is the dissipative force, and $\mathbf{F}_{ij}^R$ is the random force.

In S-MDPD, the above three forces are defined as follows:

$$\mathbf{F}_{ij}^C = (A_{ij}\omega_c(r_{ij}) + B_{ij}(\bar{\rho}_i + \bar{\rho}_j)\omega_d(r_{ij}))\mathbf{e}_{ij}, \quad (2)$$

$$\mathbf{F}_{ij}^D = -\gamma^\parallel \omega_D^\parallel(r_{ij})(\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij}, \quad (3)$$

$$\mathbf{F}_{ij}^R = \sigma^\parallel \omega_R^\parallel(r_{ij})\Delta t^{-1/2}(\mathbf{e}_{ij} \cdot \boldsymbol{\theta}_{ij})\mathbf{e}_{ij}, \quad (4)$$

where r_{ij} is the distance between particles i and j , and $\mathbf{e}_{ij} = (\mathbf{r}_i - \mathbf{r}_j)/r_{ij}$ is the unit vector in the direction from particle j to i .

For the calculation of $\mathbf{F}_{ij}^C$, A_{ij} is the strength of the attractive force and B_{ij} is the strength of the repulsive force. Note that A is always negative to create a van der Waals loop and the liquid–vapor interface, whereas B should be positive to keep the core of the force field repulsive.¹⁰ $\omega_c(r) = (1 - r/r_c)$ and $\omega_d(r) = (1 - r/r_d)$ are weight functions that vanish for $r > r_c$ and $r > r_d$, respectively. $\bar{\rho}_i = \sum_{j \neq i} \omega_\rho(r_{ij})$ is the local density for each particle within an effective range of r_d , where $\omega_\rho(r) = 105/(16\pi r_d^3)(1 + 3r/r_d)(1 - r/r_d)^3$ is the weight function.

In Eq. (3), $\gamma^\parallel$ represents the strength of the dissipative force, also called the friction coefficient. $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the velocity difference, where $\mathbf{v}_i$ and $\mathbf{v}_j$ are the velocities of particles i and j . The weight function is $\omega_D^\parallel(r) = (1 - r/r_c)^2$, which has an interaction range of r_c .

In Eq. (4), $\sigma^\parallel$ is the strength of the random force, also called the noise coefficient. Moreover, $(\sigma^\parallel)^2 = 2\gamma^\parallel k_B T$, where k_B is the Boltzmann constant and T is the temperature of the system. The weight function $\omega_R^\parallel(r)$ is chosen such that $\omega_D^\parallel(r) = [\omega_R^\parallel(r)]^2$. Δt is the simulation time step. $\boldsymbol{\theta}_{ij}$ is a three-dimensional time-variant vector whose elements are independent Gaussian random numbers with zero mean and unit variance. This vector satisfies the following conditions:

$$\langle \boldsymbol{\theta}_{ij} \rangle = 0, \quad (5)$$

$$\boldsymbol{\theta}_{ij}(t) \otimes \boldsymbol{\theta}_{kl}(t') = \mathbf{I}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk})\delta(t - t'), \quad (6)$$

where $\mathbf{I}$ is a 3×3 matrix and $\otimes$ represents the dyad product.

However, the standard DPD scheme cannot be used to control the liquid properties, such as viscosity, efficiently.³⁶ Therefore, transverse effects are introduced to $\mathbf{F}_{ij}^D$ and $\mathbf{F}_{ij}^R$ as follows:

$$\mathbf{F}_{ij}^D = -\gamma^\parallel \omega_D^\parallel(r_{ij})(\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij} - \gamma^\perp \omega_D^\perp(r_{ij})[\mathbf{v}_{ij} - (\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij}], \quad (7)$$

$$\mathbf{F}_{ij}^R = \sigma^\parallel \omega_R^\parallel(r_{ij})(\mathbf{e}_{ij} \cdot \boldsymbol{\theta}_{ij})\mathbf{e}_{ij} + \sigma^\perp \omega_R^\perp(r_{ij})[\boldsymbol{\theta}_{ij} - (\mathbf{e}_{ij} \cdot \boldsymbol{\theta}_{ij})\mathbf{e}_{ij}]. \quad (8)$$

Here, $\gamma^\perp$ and $\sigma^\perp$ are perpendicular to the direction of $\mathbf{e}_{ij}$, and $(\sigma^\perp)^2 = 2\gamma^\perp k_B T$. Additionally, $\omega_D^\perp$ and $\omega_R^\perp$ are the corresponding weight functions with the same cutoff radii and relation as between $\omega_D^\parallel$ and $\omega_R^\parallel$. In this T-MDPD scheme, $\mathbf{F}_{ij}^C$ is the same as in the standard scheme. When $\gamma^\perp = \sigma^\perp = 0$, we recover the standard scheme.

B. Simulation setup for calculation of the viscosity

Although kinetic theory can be used to calculate the dynamic viscosity μ , the viscosity obtained from DPD simulations generally varies

from this theoretical prediction.^{39,40} To date, several methods have been used as numerical viscometers, such as Couette flow, Poiseuille flow, nonequilibrium molecular dynamics, and so on.^{35,41–45} In current MDPD simulations, the Poiseuille flow method is used to calculate the viscosity, and the simulation setup is presented in Fig. 1.

The dimensions of the simulation box are L_{x1} in the x -direction, $L_{y1} = 60.0$ in the y -direction, and $L_{z1} = 8.0$ in the z -direction. A Newtonian liquid extending from $L_{yp1} = 2.77$ to $L_{yp2} = 33.28$ is confined between two solid walls in the x - z plane. The particle number density of the lower and upper walls is 12.0, and these walls consist of 8352 and 9396 solid particles, respectively. For original liquid particle number densities of 4.0, 5.0, 6.0, 7.0, 8.0, and 9.0, the system consists of 29 2780, 38 052, 45 229, 52 560, 60 648, and 65 531 liquid particles, respectively.

In the simulation, uniform body forces are applied to all of the liquid particles in the x -direction to generate a Poiseuille flow, as both solid walls are stationary during the simulation. In addition, periodic boundary conditions are applied along the x - and z -directions. After flow equilibration, the simulation box is divided into $N_{s1} = 600$ thin layers along the y -direction, and the velocity U_x^p in the x -direction is averaged within every layer.

According to the analysis of a Poiseuille flow, the averaged velocity profile can be fitted with a parabolic function,³⁹

$$U_x^p = \frac{\rho f_x}{2\mu} (H_p y - y^2) + u_s, \quad (9)$$

where H_p is the distance between the two plates, ρ is the particle number density after simulation equilibration, f_x is the body force, and u_s represents the slip velocity at the solid walls. $y = 0$ is the location of the topmost layer of the lower plate. Based on the second-order coefficient of the fitting function, μ can be obtained for a given series of input parameters.

In the simulations, the temperature is calculated by the formula $KE = DOF/2k_B T$, where KE is the total kinetic energy of all particles assigned to a given layer and DOF is the total number of degrees of

freedom for those particles. The temperature in the given layer can then be acquired as⁴²

$$(k_B T)_i = m_i v_{zi}^2, \quad (10)$$

where v_{zi} is the average velocity in the z -direction within the i th layer and m_i is the mass of the i th layer. Here, the velocity of the particle along the x - and y -directions is the same as the velocity in the z -direction.

C. Parameters used in the simulations

With the addition of a modified MDPD class, all the simulations are executed using the particle dynamics software code LAMMPS.⁴⁶ The interaction parameters are listed in Table I in reduced units, which are representative values in an MDPD system.²¹

In the simulation of a Poiseuille flow, the number of time steps for the initial thermo-equilibration is 5.0×10^4 . After the thermo-equilibration, the liquid particles in the channel are driven by a steady external force f_x in the x -direction. After 1.0×10^5 time steps, a steady flow is developed, and the simulation box is divided into thin layers along the y -direction so that the density, velocity, and temperature profiles can be calculated. These quantities are obtained by averaging over 1.0×10^5 iterations. A Poiseuille flow with a simulation box length of $L_{x1} = 30.0$ is developed by applying an external force with a value of $f_x = 0.01$. The profiles of the velocity and temperature obtained from the S-MDPD and T-MDPD schemes are compared in Fig. 2.

For the S-MDPD and T-MDPD schemes where the transverse friction coefficient $\gamma^\perp = 0$, the velocity distributions are almost the same. The velocities are higher than those obtained from the T-MDPD scheme with $\gamma^\parallel = \gamma^\perp = 18$. This not only reveals the reliability of the T-MDPD scheme, but also indicates the influence of the transverse coefficient on the liquid property, which will be discussed in Sec. III. As shown in Fig. 2(b), either the S-MDPD thermostat or the T-MDPD thermostat can maintain the temperature at a steady value of $k_B T = 1$. Moreover, the particle number density is about $\rho = 6.27$ across the channel, except for the near-wall oscillations.

The flow properties are also compared in Fig. 3 for various values of f_x and L_{x1} of a T-MDPD fluid. The results show that $f_x = 0.01$ is sufficient to maintain the temperature at the target value of $k_B T = 1.0$, while an obvious deviation from the target value is observed at higher driving forces, such as $f_x = 0.2$. At L_{x1} values of

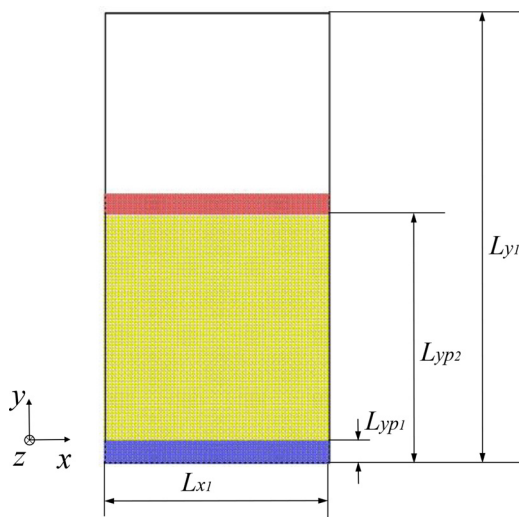


FIG. 1. Front view of the simulation domain for calculation of the viscosity.

TABLE I. Parameters (in DPD units) used in current simulations.

Parameter	Symbol	Value
Attractive coefficient	A	-4.0
Repulsive coefficient	B	25.0
Cutoff radius of attractive force	r_c	1.00
Cutoff radius of repulsive force	r_d	0.75
Friction coefficient	γ	18.0
Random coefficient	σ	6.0
System temperature	$k_B T$	1.0
Time step	Δt	0.01

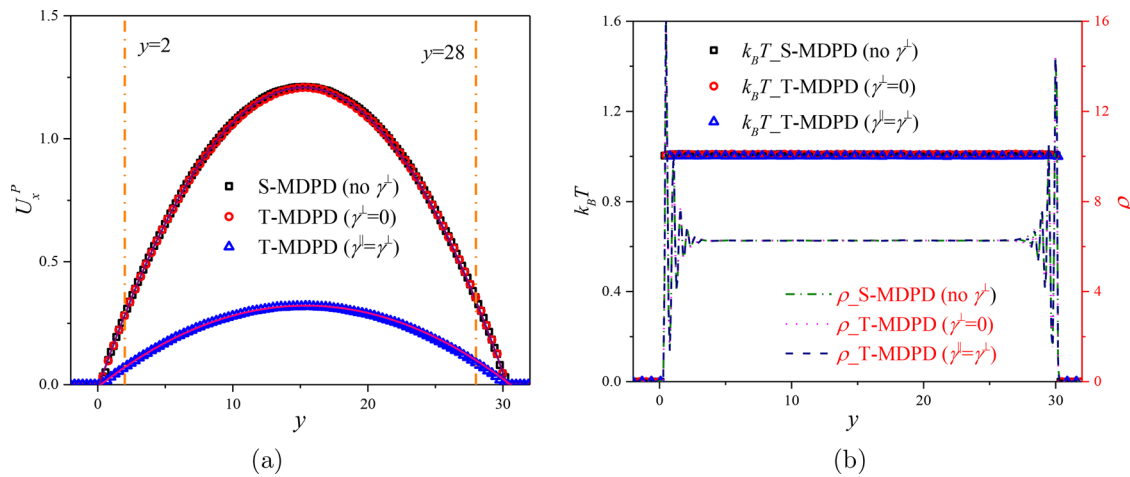


FIG. 2. Velocity (a) and temperature and density (b) distributions across the channel ($\rho = 6.27$). The friction coefficients are $\gamma^\parallel = 18$ for S-MDPD (no $\gamma^\perp$), $\gamma^\parallel = 18$ and $\gamma^\perp = 0$ for T-MDPD ($\gamma^\perp = 0$), and $\gamma^\parallel = \gamma^\perp = 18$ for T-MDPD ($\gamma^\parallel = \gamma^\perp$), respectively. In (a), the solid lines are fitting curves based on the velocity profile and the dot-dashed lines represent the fitting range.

$15.0r_c$, $30.0r_c$, and $60.0r_c$ (the default value of $r_c = 1.0$), the corresponding liquid particle numbers are 4752, 45 229, and 90 459, respectively, and the temperature remains almost steady at a value of around 1.0.

The viscosity values in these cases are listed in Table II. The viscosity is about 22.0 for all driving forces and simulation box lengths. This not only reveals that the fluid is in the Newtonian regime, but also indicates that the current simulated system size has little impact on the liquid properties. Thus, values of $f_x = 0.01$ and $L_{x1} = 30.0$ are used for subsequent simulations of the Poiseuille flow.

In the current study, the friction coefficients $\gamma^\parallel$ and $\gamma^\perp$, the attractive coefficient A , the repulsive coefficient B , and the cutoff radii r_c and r_d are all varied to investigate the effects on the viscosity for an MDPD fluid. Unless specifically noted, the viscosity values are all obtained from the T-MDPD scheme with the representative

values given in Table I, and the temperature and the liquid particle number density remain steady across the channel after the flow reaches equilibrium. Additionally, the default setting for a T-MDPD fluid is $\gamma^\parallel = \gamma^\perp$.

III. RESULTS AND DISCUSSION

A. Effect of the transverse coefficient

First, the viscosity obtained from T-MDPD and S-MDPD schemes is compared in Fig. 4. For a T-MDPD fluid with a transverse friction coefficient of $\gamma^\perp = 0$, the viscosity is equal to that obtained from the S-MDPD scheme with the same value of the parallel friction coefficient $\gamma^\parallel$, which corresponds to the similar velocity profiles shown in Fig. 2(a). However, when $\gamma^\parallel$ and $\gamma^\perp$ are both active in the T-MDPD scheme, the viscosity is much higher than in the S-MDPD fluid.

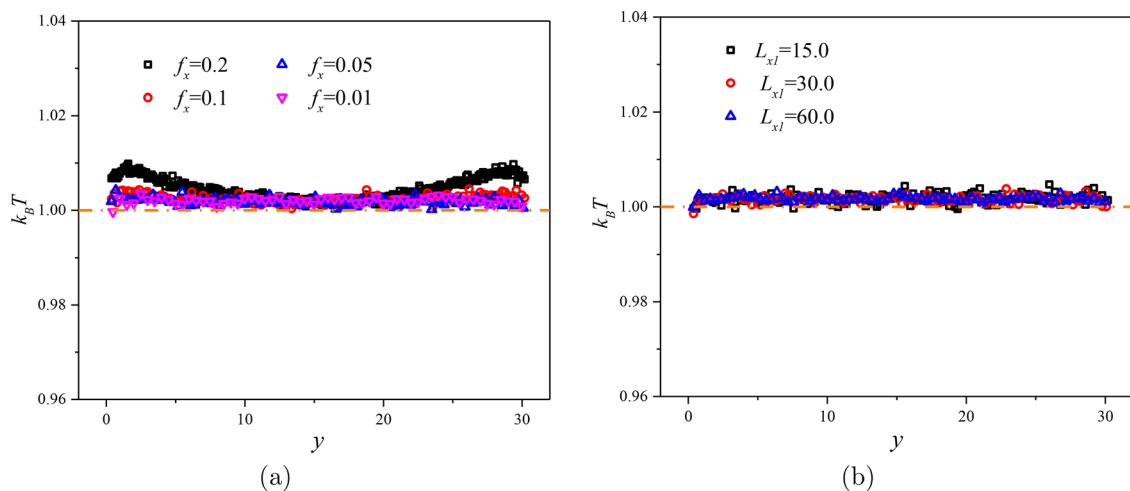


FIG. 3. Temperature profiles across the channel for various values of f_x (a) and L_{x1} (b) in a T-MDPD fluid ($\rho = 6.27$). $L_{x1} = 30.0$ in (a), $f_x = 0.01$ in (b).

TABLE II. Viscosity μ of a T-MDPD fluid for various driving forces f_x and simulation box lengths L_{x1} .

f_x	μ	L_{x1}	μ
0.10	21.71	15.0	23.06
0.05	22.15	30.0	22.68
0.01	22.68	15.0	22.24
$L_{x1} = 30.0$		$f_x = 0.01$	

For example, the viscosity of the T-MDPD fluid increases from 7.50 to 49.02 as γ increases from 4.5 to 40.5 in DPD units, whereas that of the S-MDPD fluid only increases from 3.77 to 9.79. Specifically, the viscosity of the T-MDPD fluid is 3.73 times that of the S-MDPD fluid for $\gamma = 18.0$, and the ratio increases to 5.0 as γ increases to 40.5. This means that the transverse effect plays an important role in the viscosity of the T-MDPD liquid, and the viscosity is highly sensitive to the friction coefficient.

In previous studies with a classical MDPD scheme, where only parallel coefficients act in the calculation of the dissipative force and the random force, kinetic theory is used to predict the viscosity in the form^{40,42,47}

$$\mu = \frac{45k_B T}{4\pi\gamma r_c^3} + \frac{2\pi\rho^2\gamma r_c^5}{105(d^2 + 2d)k_B T}, \quad (11)$$

where d is the dimensionality of the simulation system and the two terms on the right-hand side arise from the kinetic contribution and the dissipative contribution, respectively. The theoretical viscosity obtained from Eq. (11) is plotted as a green dashed line in Fig. 4. It can be seen that these theoretical values are always lower than the viscosity obtained from the S-MDPD simulation, as observed in other research.^{39,42}

Due to the relatively low temperature of $k_B T = 1.0$ in the simulation, the collision time t_w is significantly larger than the kinetic

relaxation time t_w , and so the kinetic contribution to the viscosity is always ignored.³⁴ Some researchers have pointed out that when the friction coefficient is greater than unity, the dissipative force is definitely dominant over the momentum.⁴⁰ Consequently, the viscosity of a classical DPD/MDPD fluid can be approximately predicted by the dissipative contribution as

$$\mu = \frac{2\pi\rho^2\gamma r_c^5}{105(d^2 + 2d)k_B T}, \quad (12)$$

where the variation of the viscosity is considered to be proportional to the friction coefficient γ . This is plotted in Fig. 4 with a red solid line. As the difference between the two theoretical lines is relatively small, the dominance of the dissipative contribution to the viscosity of an MDPD fluid is reasonable. The deviation between the theoretical values and the simulation results will be discussed later.

The viscosity of the S-MDPD fluid and the T-MDPD fluid varies linearly with respect to the friction coefficient, as shown by the straight dashed lines in Fig. 4. To investigate the influence of the transverse coefficient on the properties of the T-MDPD fluid, the viscosities are systematically compared for various friction coefficients in Fig. 5. The results show that the viscosity increases with increasing friction coefficient γ , and the density in the bulk region remains almost constant as γ varies.

As the transverse effect is introduced into the T-MDPD fluid, the viscosity becomes much higher than that obtained from the S-MDPD scheme, which makes the classical theory unsuitable. However, note that the viscosity also varies linearly with respect to γ , which can be fitted with a linear function, as shown by the dotted and dashed lines in Fig. 5(a). This means that the viscosity is also proportional to the friction coefficient γ in the T-MDPD fluid. The fitting functions will be given later.

Actually, the viscosity of a DPD/MDPD fluid not only consists of kinetic and dissipative parts, but also includes a conservative part. Although this contribution is often neglected in the derivation of transport properties, some researchers have indicated that the conservative force is important in modeling a DPD/MDPD fluid.^{48–50} Additionally, Zhao *et al.* have observed that the contribution of the conservative force to the viscosity increases considerably with increasing repulsive coefficient, density, and cutoff radius in an MDPD model.³⁴

Let us now return to the deviation between the theoretical values given by Eq. (12) [or Eq. (11)] and the simulation results for the S-MDPD fluid. Coupled with the above analysis of the viscosity components of an MDPD fluid, this deviation can be considered as the conservative contribution. The dependency of the viscosity on the conservative force parameters, such as the liquid particle number, can be observed in Fig. 5. The dependency of the viscosity on the conservative force parameters for the T-MDPD fluid is explored in Sec. III B.

B. Effect of the conservative force parameters

As the viscosity of the T-MDPD fluid is sensitive to the liquid particle number density, as shown in Fig. 5(a), the influence of the density on the viscosity is explored in Fig. 6. For various friction coefficients, it can be seen that the viscosity definitely increases with increasing particle number density. The dashed lines illustrate the best fit to

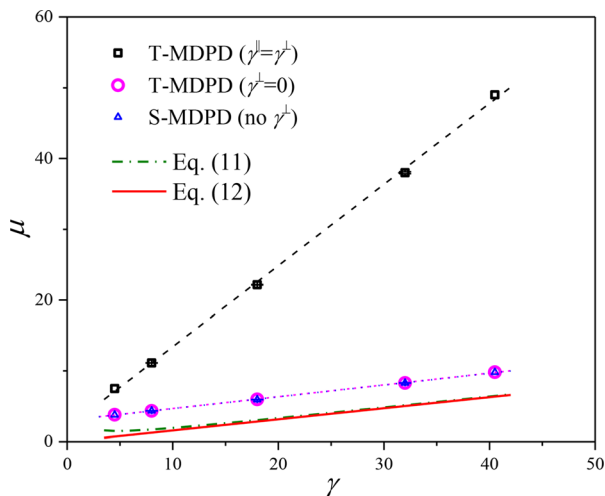


FIG. 4. Variation of the viscosity with increasing friction coefficient γ ($\rho = 6.27$). The x-axis represents both $\gamma^{\parallel}$ and $\gamma^{\perp}$ for “T-MDPD ($\gamma^{\parallel} = \gamma^{\perp}$),” whereas it only represents $\gamma^{\parallel}$ for “T-MDPD ($\gamma^{\perp} = 0$)” and “S-MDPD (no $\gamma^{\perp}$).” The error bars are the standard errors in the mean values evaluated for each point.

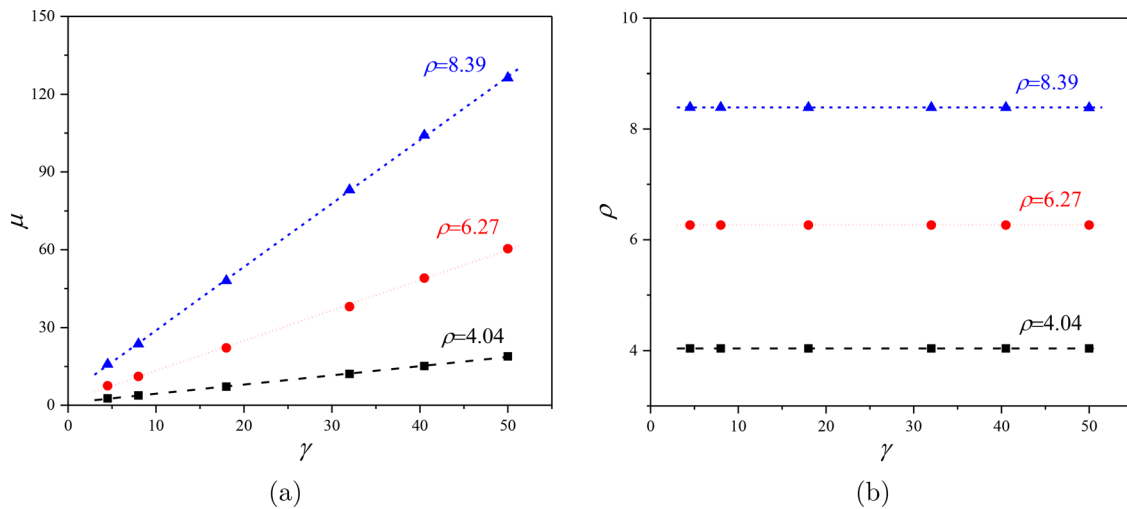


FIG. 5. Variation of (a) viscosity and (b) liquid particle number density in the bulk region with increasing diffusion coefficient γ for various densities.

the simulation values. The fitting function, as well as the other fitting functions used in Figs. 7 and 8 are discussed in Sec. III C.

Previous studies have shown that the cutoff radii are essential in calculating the viscosity of a DPD/MDPD fluid. Specifically, the cutoff radius r_c affects the conservative interactions as well as the dissipative and random interactions. For the T-MDPD fluid, the results show that the viscosity increases with increasing r_c . Although Abouzar *et al.* stated that increasing r_c results in an achievable upper limit for the viscosity at which T can be effectively controlled,⁵¹ the temperature remains at about $k_B T = 1.0$ for all r_c in the current simulations.

The cutoff radius r_d affects the calculation of the conservative force. The viscosity obtained with different values of r_d is shown by the

open symbols in Fig. 7. It can be seen that the viscosity changes slightly as r_d varies, i.e., r_d has a much smaller influence on the liquid viscosity than r_c . Consequently, the cutoff radius r_d can be neglected in the prediction of the viscosity.

Besides the liquid particle number density and cutoff radii, the repulsive coefficient and attractive coefficient are essential components of the conservative force. Here, the dependency of the viscosity on the repulsive coefficient B is investigated for various densities and attractive coefficients. The results show that increasing B also increases the viscosity of the T-MDPD fluid. Moreover, larger values of the density produce higher viscosity values, as shown in Fig. 8(a), which is consistent with Fig. 6.

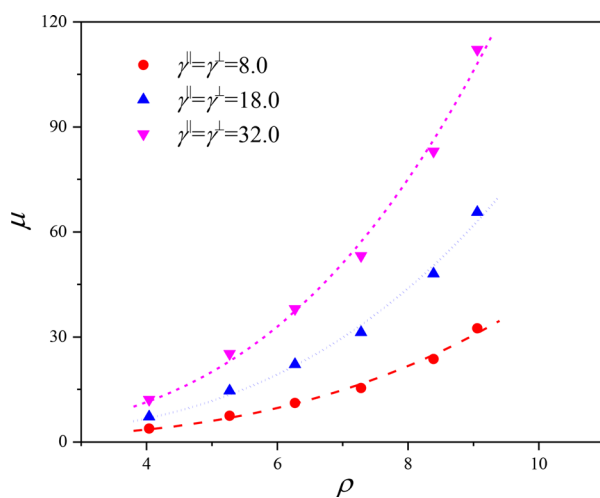


FIG. 6. Variation of the viscosity with increasing liquid particle number density ρ for various friction coefficients.

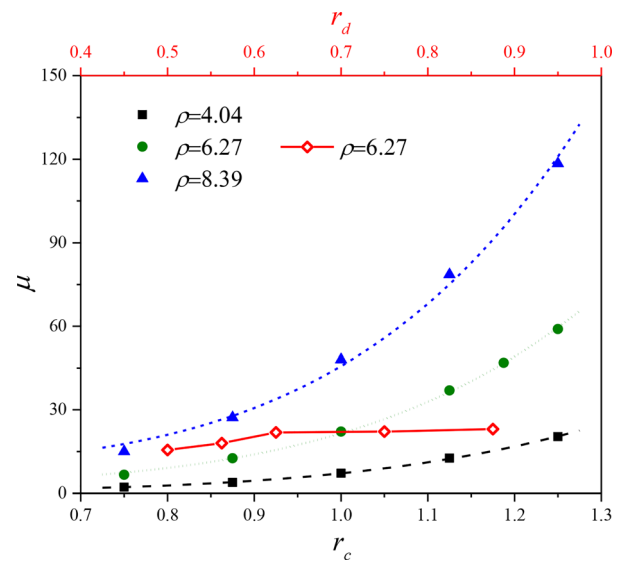


FIG. 7. Variation of the viscosity with increasing cutoff radii r_c (bottom x-axis) and r_d (top x-axis).

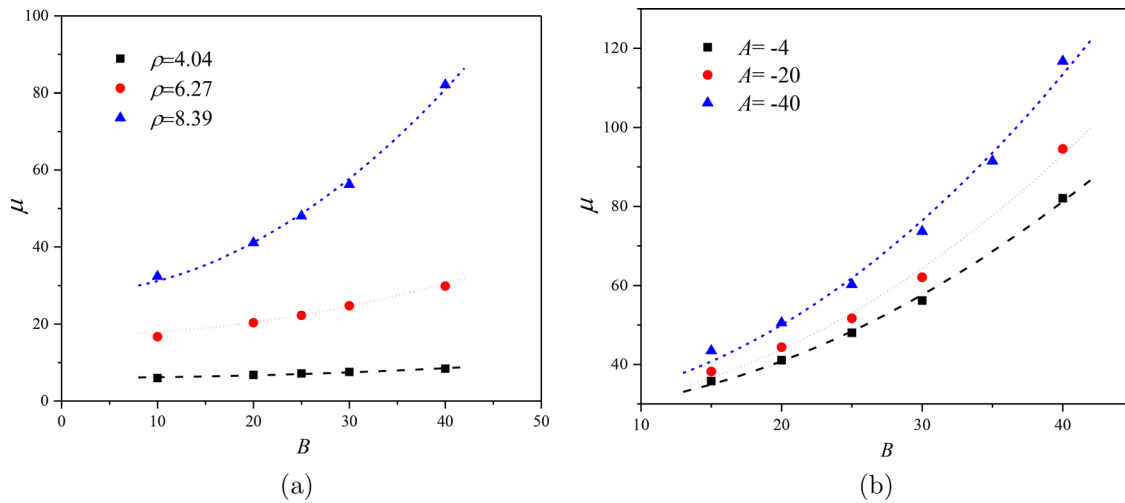


FIG. 8. Variation of the viscosity with increasing repulsive coefficient B for various densities (a) and attractive coefficients ($\rho = 8.39$) (b).

C. Empirical equation for predicting the viscosity

In the simulations of a viscometric flow for an MDPD fluid, Zhao *et al.* have suggested a simple fitting equation for predicting the conservative viscosity, and their results show that this is able to match the conservative viscosity reasonably well.³⁴ This equation shows that the conservative viscosity is related to the repulsive coefficient and the liquid particle number density. As the dependency of the viscosity on these conservative parameters was also observed in Sec. III B, this equation should be able to predict the conservative part for the current T-MDPD fluid. Additionally, the viscosity varies linearly with respect to the friction coefficient, and the trend is similar to Eq. (12). Consequently, by neglecting the kinetic contribution, the viscosity equation for the T-MDPD fluid can be simplified to

$$\mu = a\rho^b B^2 + c\rho^2 \gamma r_c^5, \quad (13)$$

where the terms on the right-hand side represent the conservative contribution and the dissipative contribution, respectively; a , b , and c are the fitting parameters, which vary for different input parameters.

According to Eq. (13), the variation of the viscosity can be described by a polynomial. Specifically, take the results shown in Fig. 6 as an example. We vary the liquid particle number density ρ and fix the other parameters (i.e., the repulsive coefficient B , the cutoff radius r_c and the friction coefficient γ). Then, the variation of the viscosity can be fitted with a curve of the form $a_1 \rho^{b_1} + c_1 \rho^2$, as shown by the dashed lines in Fig. 6. The fitting precision of the simulation data approaches 0.99 for all three friction coefficients. For given input parameters, the fitting parameters are listed in Table III.

With the exception of some isolated cases, the fitting precision to the simulation values approaches 0.95. This means that whereas the fitting parameters vary for different input parameters, Eq. (13) can be used to qualitatively describe the variation of the viscosity in the T-MDPD fluid. However, the attractive coefficient A is not included in Eq. (13), but A clearly influences the viscosity, as can be seen from Fig. 8(b). Indeed, the dependency of the viscosity on A has been observed in previous studies, but the effect of A has not been considered in the derivation of the conservative viscosity.³⁴ Here, the influence of A is investigated for various repulsive coefficients with a fixed liquid particle number density of 8.39.

The results show that the viscosity decreases with increasing A , which means that the liquid becomes less viscous. In addition, the viscosity is basically quadratic in the attractive coefficient A for a liquid particle number density of $\rho = 8.39$, and the simulation values fit a curve of the form $a_5 A^2 + c_5$ well (the fitting parameters are listed in Table III). However, once the original liquid particle number decreases to 6.0 or less, or the attractive coefficient A becomes as low as -20 , the Poiseuille flow cannot be formed. Unless the upper wall is displaced, which will influence the density in the bulk region, the flow contains a cavity as the liquid particles accumulate together.

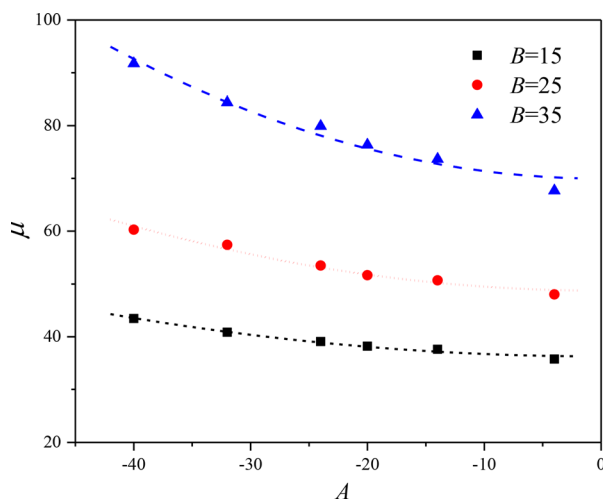


FIG. 9. Variation of the viscosity with increasing attractive coefficient A for various repulsive coefficients ($\rho = 8.39$).

TABLE III. Fitting parameters for the variation of the viscosity in the T-MDPD fluid.

Figures	Fitting functions	Input parameters	Fitting parameters	Adj. R ²
Fig. 5(a)	$a_1 + c_1\gamma$	$\rho = 4.04$	$a_{11} = 0.890, c_{11} = 0.355$	0.999
		$\rho = 6.27$	$a_{12} = 1.697, c_{12} = 1.163$	0.999
		$\rho = 8.39$	$a_{13} = 4.429, c_{13} = 2.448$	0.999
Fig. 6	$a_2\rho^{b_2} + c_2\rho^2$	$\gamma = 8$	$a_{21} = 0.877, b_{21} = 4.501 \times 10^{-18}, c_{21} = 0.041$	0.985
		$\gamma = 18$	$a_{22} = 1.317, b_{22} = 9.197 \times 10^{-18}, c_{22} = 0.083$	0.988
		$\gamma = 32$	$a_{23} = 2.239, b_{23} = 15.753 \times 10^{-18}, c_{23} = 1.423$	0.989
Fig. 7	$a_3 + c_3r_c^5$	$\rho = 4.04$	$a_{31} = 0.635, c_{31} = 6.501$	0.999
		$\rho = 6.27$	$a_{32} = 3.010, c_{32} = 18.532$	0.999
		$\rho = 8.39$	$a_{33} = 8.997, c_{33} = 36.690$	0.994
Fig. 8(a)	$a_4B^2 + c_4$	$\rho = 4.04$	$a_{41} = 1.56 \times 10^{-3}, c_{41} = 6.047$	0.950
		$\rho = 6.27$	$a_{42} = 8.44 \times 10^{-3}, c_{42} = 17.077$	0.923
		$\rho = 8.39$	$a_{43} = 33.18 \times 10^{-3}, c_{43} = 27.897$	0.995
Fig. 9	$a_5A^2 + c_5$	$B = 15$	$a_{51} = 4.55 \times 10^{-3}, c_{51} = 36.274$	0.979
		$B = 25$	$a_{52} = 7.62 \times 10^{-3}, c_{52} = 48.752$	0.973
		$B = 35$	$a_{53} = 14.15 \times 10^{-3}, c_{53} = 69.968$	0.959

Consequently, the variation of the viscosity with the attractive coefficient is not included in Eq. (13) for general T-MDPD fluids. For heavier fluids, such as with a particle number density of $\rho = 8.39$, the equation used to predict the viscosity of a T-MDPD fluid can be reformulated as

$$\mu = a\rho^b B^2 A^2 + c\rho^2 \gamma r_c^5. \quad (14)$$

Note that Eq. (14) is only suitable for heavier fluids. In experimental research, aqueous glycerol solution is often used to adjust the viscosity. In numerical studies, higher viscosity values can be obtained by increasing the liquid particle number density ρ , the friction coefficient γ , the repulsive coefficient B , the cutoff radius r_c , and the absolute value of the attractive coefficient A . Lower viscosities can be obtained by decreasing ρ , γ , B , and r_c .

For example, for input parameters of $\rho = 8.39$, $\gamma^{\parallel} = \gamma^{\perp} = 50$, $B = 40$, $r_c = 1.25$, and $A = -40$ the viscosity μ for a T-MDPD fluid is 270.17 in DPD units. The viscosity μ is only 1.13 in DPD units when the input parameters are $\rho = 4.04$, $\gamma^{\parallel} = \gamma^{\perp} = 4.5$, $B = 10$, $r_c = 0.75$, and $A = -4$. The ratio of the viscosity between these two T-MDPD fluids is similar to the ratio of 90% aqueous glycerol solution to water at a temperature of 20 °C. As this T-MDPD scheme can be used to effectively tune the fluid viscosity, it will be useful in studies of multiphase flows, such as adjusting the droplet impact behaviors on solid or liquid surfaces, or modifying the flow slip characteristics over liquid-liquid interfaces as mentioned in Sec. I.

IV. CONCLUSIONS

In S-MDPD equations, the pairwise forces consisting of the conservative force, the dissipative force, and the random force only act on the relative velocity along the interparticle axis. In this paper, we have proposed a transverse MDPD scheme in which a perpendicular friction coefficient is added to the dissipative force and random force. A Poiseuille flow method was applied as a numerical viscometer to explore the viscosity of the resulting T-MDPD fluid.

When a lateral effect is introduced to the MDPD equations, the viscosity of the T-MDPD fluid is much higher than that of an S-MDPD fluid, and the viscosity is sensitive to the new transverse friction coefficient. Increasing the friction coefficient increases the viscosity of the T-MDPD fluid, as does increasing conservative force parameters such as the liquid particle number density, cutoff radius, and repulsive coefficient. Based on this extension of the MDPD scheme, the viscosity of an MDPD fluid can be adjusted over a large range. Finally, an empirical equation was proposed that provides quantitative predictions of the viscosity of the T-MDPD fluid. As the fluid viscosity plays an important role in multiphase flows, this T-MDPD scheme will be of great value in future studies of such flows using the MDPD simulations.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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